

ADAR1000-EVALZ ↔ ZC702 FPGA

SPI Interface Setup Guide

PMOD J63 (ZC702) → P3 PMOD (ADAR1000-EVALZ) · RX1 Maximum Gain · 45° Phase Shift

This document provides a complete step-by-step guide to connecting the **ADAR1000-EVALZ** 4-channel X/Ku-band beamformer board to the **Xilinx ZC702** evaluation board (XC7Z020) using the **PMOD J63** connector and the **P3 PMOD** header on the ADAR1000-EVALZ. It covers the hardware wiring, SPI protocol, the 12-step **RX1_MaxG_45** register-write sequence, FPGA implementation flow, and expected VNA measurement results.

1. Hardware Required

Item	Qty	Notes
Xilinx ZC702 Evaluation Board	1	XC7Z020-CLG484-1
ADAR1000-EVALZ Board	1	Analog Devices beamformer eval board
6-wire jumper cable (F-F)	1	Female-to-female, ≤150 mm
External +3.3 V bench supply	1	AVDD3 rail, ≤500 mA
External -5.0 V bench supply	1	AVDD1 rail, ≤200 mA
USB micro-B cable	1	ZC702 JTAG / UART
Vivado 2022.x or later	—	Free WebPACK edition sufficient
VNA (optional)	1	8–16 GHz for RF verification

■ The -5 V supply cannot be obtained from the ZC702 or the PMOD connector. A bench power supply or dedicated dc-dc converter is required.

2. Wiring Diagram

Connect the six wires between **ZC702 PMOD J63** and **ADAR1000-EVALZ P3** as shown in the diagram below. Always connect GND first and disconnect power before making changes.

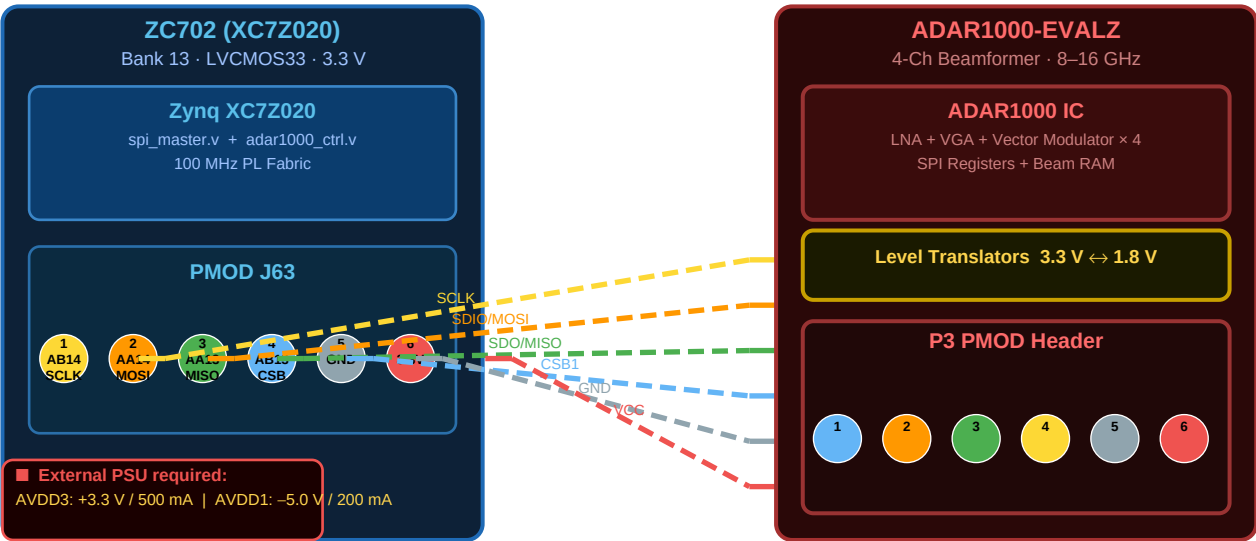


Figure 1 — Hardware wiring diagram: ZC702 J63 ↔ ADAR1000-EVALZ P3

2.1 Wire-by-Wire Connection Table

#	Colour	ZC702 J63 Pin	FPGA Pin	ADAR1000 P3 Pin	Signal
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1	Yellow	Pin 1	AB14	Pin 4	SCLK — SPI clock
2	Orange	Pin 2	AA14	Pin 2	SDIO — MOSI (FPGA→ADAR1000)
3	Green	Pin 3	AA13	Pin 3	SDO — MISO (ADAR1000→FPGA)
4	Blue	Pin 4	AB13	Pin 1	CSB1 — Chip Select (active-low)
5	Black	Pin 5	GND	Pin 5	Ground
6	Red	Pin 6	3.3 V	Pin 6	VCC — Logic supply

3. Step-by-Step Connection Procedure

Step 1 — Power OFF

Turn off both boards and all bench supplies before connecting any wires.

Step 2 — Ground first

Connect J63 Pin 5 (GND) → P3 Pin 5 (GND) with the black wire. This establishes a common reference before signal wires are attached.

Step 3 — SPI signals

Connect the four SPI wires: SCLK (Yellow J63-1→P3-4), SDIO/MOSI (Orange J63-2→P3-2), SDO/MISO (Green J63-3→P3-3), CSB1 (Blue J63-4→P3-1).

Step 4 — Logic VCC

Connect J63 Pin 6 (3.3 V) → P3 Pin 6 (VCC) with the red wire. This powers the on-board level translators (3.3 V ↔ 1.8 V).

Step 5 — External power

Connect bench supplies: +3.3 V → AVDD3 (≤500 mA) and –5.0 V → AVDD1 (≤200 mA). Set current limits before applying power.

Step 6 — Program FPGA

Open Vivado, create an RTL project with rtl/*.v and constraints/zc702_pmod_j63.xdc. Set SPI_CLK_DIV=4 (SCLK=12.5 MHz), RESET_WAIT_CYCLES=10000 (100 μs). Run Synthesis → Implementation → Generate Bitstream → Program Device.

Step 7 — Power-on sequence

Apply AVDD3 and AVDD1 first. Then power on the ZC702 (12 V supply). Wait for the DONE LED. The ADAR1000 is configured automatically; led_done_o illuminates when all 12 SPI writes are complete.

Step 8 — VNA verification (optional)

Connect a VNA between RF_IO and RX1 SMAs. Sweep 8–16 GHz. Expected: gain ≈ +10 dB, return loss < –10 dB, phase shift = 45° on Ch1.

4. SPI Protocol Details

4.1 24-bit SPI Word Format

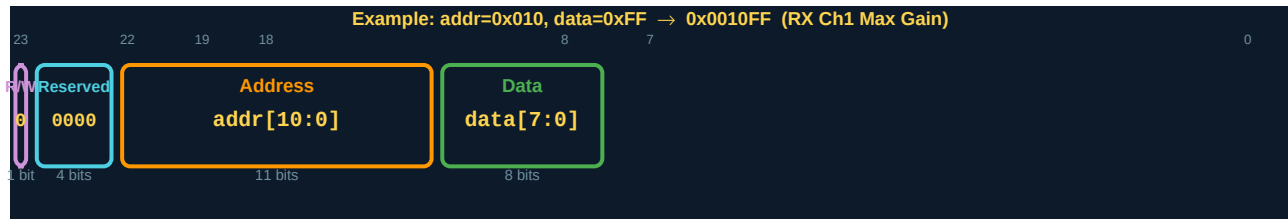


Figure 2 — 24-bit SPI word layout: {R/W[23], Reserved[22:19], Addr[18:8], Data[7:0]}

Parameter	Value
Transaction width	24 bits
Bit order	MSB first (bit 23 → bit 0)
SPI mode	Mode 0 (CPOL=0, CPHA=0)
Data sampled on	Rising SCLK edge
Data shifted on	Falling SCLK edge
Max SCLK (ADAR1000)	50 MHz
SCLK in this design	12.5 MHz (CLK_DIV=4, 100 MHz system clock)
CSB polarity	Active-low
Logic levels	3.3 V (FPGA) ↔ 1.8 V (ADAR1000 core, via level translators)

4.2 SPI Timing Diagram

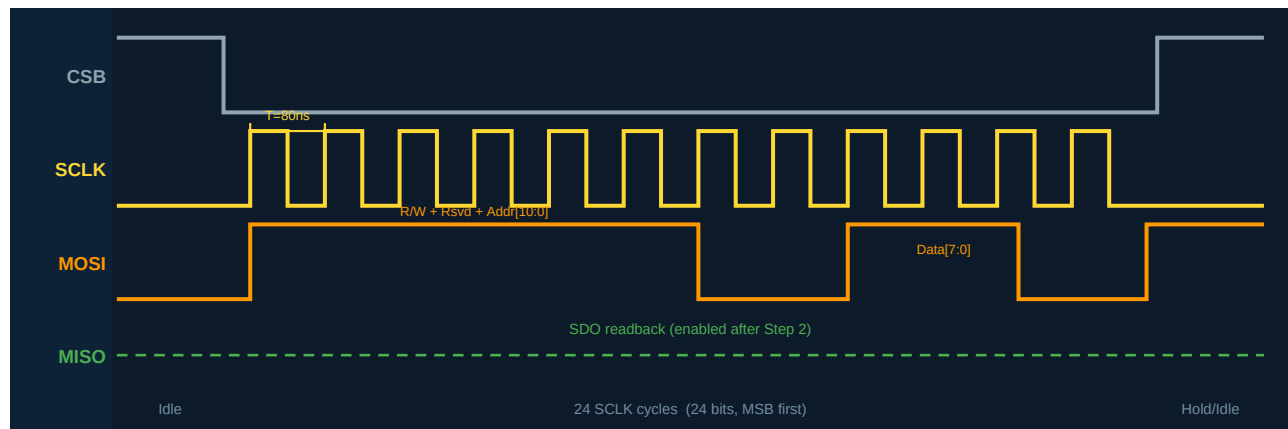


Figure 3 — SPI Mode 0 timing: CSB low → 24 SCLK cycles → CSB high. MOSI valid before rising edge; MISO sampled on rising edge.

5. RX1_MaxG_45 Register Sequence

The following 12 SPI write transactions configure the ADAR1000 for **RX Channel 1 at maximum gain with a 45° phase shift**. The sequence must be executed in the order shown. After Step 1 (Soft Reset) a mandatory delay of $\geq 100\ \mu\text{s}$ must elapse before Step 2.

Step	SPI Word	Reg	Data	Description	Section
1	0x000081	0x000	0x81	Soft Reset	Init & Reset
2	0x000018	0x000	0x18	Enable SDO, 4-wire SPI mode	Init & Reset
3	0x040055	0x400	0x55	Trim LDO to ≈1.8 V	Init & Reset
4	0x003860	0x038	0x60	RAM Bypass — working registers	RAM Bypass
5	0x002E7F	0x02E	0x7F	RX Enable — all 4 channels	RAM Bypass
6	0x003408	0x034	0x08	LNA Bias	Bias & Switch
7	0x003516	0x035	0x16	RX VGA / VM Bias	Bias & Switch
8	0x003120	0x031	0x20	TR Switch → Receive mode	Bias & Switch
9	0x0010FF	0x010	0xFF	RX Ch1 gain = maximum	Ch1 Config
10	0x001436	0x014	0x36	RX Ch1 VM I = 0x36 (45°)	Ch1 Config
11	0x001536	0x015	0x36	RX Ch1 VM Q = 0x36 (45°)	Ch1 Config
12	0x002801	0x028	0x01	LDRX Override — commit to RF	Load/Commit

Figure 4 — RX1_MaxG_45 SPI register write sequence (12 transactions)

5.1 Section 1: Device Initialisation & Reset

- **0x000081** (Reg 0x000, Data 0x81): Soft Reset. Sets `SOFT_RESET` bit; all registers return to default. Wait ≥100 μs.
- **0x000018** (Reg 0x000, Data 0x18): Enable SDO pin; select 4-wire SPI mode. `SDO_ACTIVE=1` enables readback.
- **0x040055** (Reg 0x400, Data 0x55): LDO trim. Adjusts the internal bandgap to deliver a stable ≈1.8 V to the BiCMOS core.

5.2 Section 2: RAM Bypass & Operating Mode

- **0x003860** (Reg 0x038, Data 0x60): RAM Bypass. `BEAM_RAM_BYPASS=1`: chip reads phase/gain from working registers, not Beam RAM.
- **0x002E7F** (Reg 0x02E, Data 0x7F): Receive Enable. Enables LNA and receive signal path on all four channels.
- **0x003408** (Reg 0x034, Data 0x08): LNA Bias. Sets DC bias current for the Low-Noise Amplifiers.

5.3 Section 3: Bias and Switch Control

- **0x003516** (Reg 0x035, Data 0x16): RX Bias. Configures operating currents for VGAs and Vector Modulators.
- **0x003120** (Reg 0x031, Data 0x20): TR Switch. `TR_SPI=1` overrides the external TR pin and forces receive mode.

5.4 Section 4: Channel 1 Configuration

- **0x0010FF** (Reg 0x010, Data 0xFF): RX Ch1 Max Gain. Data=0xFF (255): attenuator fully bypassed, VGA at maximum.
- **0x001436** (Reg 0x014, Data 0x36): RX Ch1 Vector Modulator I = 0x36. In-phase component of the complex weight.
- **0x001536** (Reg 0x015, Data 0x36): RX Ch1 Vector Modulator Q = 0x36. Quadrature component. Together I=Q=0x36 produces a 45° phase rotation.
- **0x002801** (Reg 0x028, Data 0x01): LDRX Override. `LDRX_OVR=1` transfers all buffered gain/phase values to the active RF path. This is the commit step.

6. FPGA Implementation Flow (Vivado)

6.1 Create Project

Open Vivado. New Project → RTL Project. Add all files from rtl/ directory: spi_master.v, adar1000_ctrl.v, top.v. Set top.v as the top-level module.

6.2 Add Constraints

Add constraints/zc702_pmod_j63.xdc to the project. This file contains PACKAGE_PIN assignments for AB14/AA14/AA13/AB13 and I/O timing constraints.

6.3 Configure Clock

Option A (PS clock): In a Zynq block design, add Clocking Wizard, set FCLK0=100 MHz, connect to top.clk port. Option B (PL-only): Use 200 MHz diff clock U64 with IBUFDS + MMCM to generate 100 MHz.

6.4 Set Parameters

In top.v: SPI_CLK_DIV=4 → SCLK=12.5 MHz. RESET_WAIT_CYCLES=10000 → 100 μs post-reset delay at 100 MHz.

6.5 Run Flow

Run Synthesis → Implementation → Generate Bitstream. Check timing report: all paths should be easily met at 100 MHz (SCLK is only 12.5 MHz).

6.6 Program Device

Open Hardware Manager, connect to ZC702 via USB-JTAG, program the .bit file. The ADAR1000 is configured automatically. led_done_o (DS12 or assigned LED) illuminates when done.

6.7 Key XDC Constraints (zc702_pmod_j63.xdc)

```
create_clock -period 10.000 -name clk [get_ports clk]
set_property -dict { PACKAGE_PIN AB14 IOSTANDARD LVCMOS33 } [get_ports spi_sclk]
set_property -dict { PACKAGE_PIN AA14 IOSTANDARD LVCMOS33 } [get_ports spi_mosi]
set_property -dict { PACKAGE_PIN AA13 IOSTANDARD LVCMOS33 } [get_ports spi_miso]
set_property -dict { PACKAGE_PIN AB13 IOSTANDARD LVCMOS33 } [get_ports spi_csb]
```

7. Alternative: Using PMOD J62

If J63 is occupied, use J62. Replace the four PACKAGE_PIN entries in the XDC file with the J62 pin assignments below. All other constraints remain identical.

Signal	J62 FPGA Pin
spi_sclk	Y12
spi_mosi	W12
spi_miso	V12
spi_csb	V13

8. Troubleshooting

Symptom	Likely Cause	Fix
led_done never asserts	FPGA not programmed / rst_n stuck low	Check JTAG connection, re-program, confirm reset logic
No SCLK visible	Wrong J63 pin	Verify AB14 against board silk screen
SPI visible, no RF output	AVDD1 (~5 V) missing	Check bench supply polarity and current limit
Gain lower than expected	LDO trim skipped	Ensure all 12 steps execute; check spi_done signal
CSB stays low	SPI master stuck in SHIFT	Check CLK_DIV parameter; verify SCLK and done signal

Wrong phase on VNA	LDRX not executed	Confirm step 12 (0x002801) completes before measurement
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9. Expected VNA Measurement Results

After executing the full **RX1_MaxG_45** configuration sequence, connect a Vector Network Analyzer between the **RF_IO** SMA port and the **RX1** SMA port on the ADAR1000-EVALZ and sweep from 8 GHz to 16 GHz. The expected results are:

Measurement	Expected Value	Significance
Insertion Gain (S21)	≈ +10 dB at mid-band (8–16 GHz)	Validates LNA + VGA at maximum gain setting
Return Loss — RX1 (S11)	< −10 dB across most of band	Confirms good impedance matching at RX1 SMA
Return Loss — RF_IO (S22)	< −10 dB across most of band	Confirms matching at the common port
Phase — RX Channel 1	45° (relative to 0° reference)	Validates I/Q vector modulator settings 0x36/0x36

Note: The oscillatory behaviour in the measured return loss is characteristic of transmission-line effects and parasitic transitions in the evaluation board's SMA connectors and PCB traces. Maintaining return loss below −10 dB ensures reflections are minimised, preserving the phase and amplitude accuracy required for null steering.

10. Repository File Reference

File	Description
<code>rtl/spi_master.v</code>	24-bit SPI master, Mode 0, MSB-first, CLK_DIV parameter
<code>rtl/adar1000_ctrl.v</code>	12-step RX1_MaxG_45 register-write sequencer
<code>rtl/top.v</code>	Top-level: binds controller + SPI master to PMOD J63 ports
<code>sim/tb_spi_master.v</code>	SPI master unit testbench (Icarus Verilog)
<code>sim/tb_adar1000_ctrl.v</code>	Integration testbench with ADAR1000 slave model
<code>constraints/zc702_pmod_j63.xdc</code>	Vivado XDC: pin assignments + timing constraints
<code>scripts/RX1_MaxG_45.txt</code>	Annotated SPI command script (hex words)
<code>docs/wiring_diagram.md</code>	Markdown wiring guide with pin tables
<code>docs/wiring_diagram.svg</code>	SVG colour wiring diagram
<code>docs/spi_timing.svg</code>	SVG SPI Mode 0 timing diagram
<code>docs/system_block_diagram.svg</code>	SVG system block diagram
<code>docs/register_map.svg</code>	SVG SPI word format and register map
<code>docs/ADAR1000_ZC702_Setup_Guide.pdf</code>	This PDF setup guide